

Phys 20BL Week 4 Out of Lab: Speed of Sound

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1 Model of Fluid for Sound Propagation

1.1 Model Derivation

First, it's assumed that the speed is given as:

$$v = \sqrt{\frac{B}{\rho}} \quad (1)$$

Where B is the Bulk Modulus (the resistance against compression of the material), while ρ is the density.

This intuitively follows from the observation that in a relatively incompressible fluid (for instance, water vs. air), the speed is traveling faster; on the other hand, the increase in density increases the amount of material the wave needs to propagate through, which naively explains the decrease in speed.

Then, the relation of Bulk Modulus is proportional to the pressure when assuming the gas is an ideal gas:

$$B = \gamma P \quad (2)$$

Where γ is the heat ratio of gas, called Adiabatic Index.

Also, with the assumption of the gas being an ideal gas, one applies the ideal gas law:

$$PV = Nk_B T \quad (3)$$

Where V is the gas volume, N is the number of gas molecules, k_B is the Boltzmann constant, and T is the temperature in Kelvin.

With these variables, the density of the gas can be given as follow:

$$\rho = \frac{Nm}{V} \quad (4)$$

Since if m represents the average mass of molecules (mass of "unit volume"), then Nm represents the total mass of the occupied volume; hence, dividing by the total volume, one recovers the density.

By plugging in all the terms from the other equations to the first one (and using the extra derivation $\frac{N}{V} = \frac{P}{k_B T}$ by ideal gas law), one yields the following derivation:

$$v = \sqrt{\frac{B}{\rho}} = \sqrt{\frac{\gamma P}{Nm/V}} = \sqrt{\frac{\gamma P}{Pm/(k_B T)}} = \sqrt{\frac{\gamma k_B T}{m}} \quad (5)$$

1.2 Assumptions of the Model

The first assumption of the model is that the gas must be ideal gas (which under this experiment without extreme compression, it'll be a suitable approximation).

On the other hand, it's also assumed that the Bulk Modulus is always proportional to the pressure of the gas P .

2 Model Prediction of Speed of Sound

2.1 Constant Values

First, here are some given constants: $k_B := 1.380649 \cdot 10^{-23} \text{ J}\cdot\text{K}^{-1}$, while $\gamma := 1.4$.

For temperature, the weather report at Goleta during the time of experiment is used, which provides $T = 16^\circ\text{C} = 289.15 \text{ K}$. In this lab, because the equipment used to observe environmental temperature / humidity / CO_2 concentration are all broken (or non-existent), one needs external sources for the data.

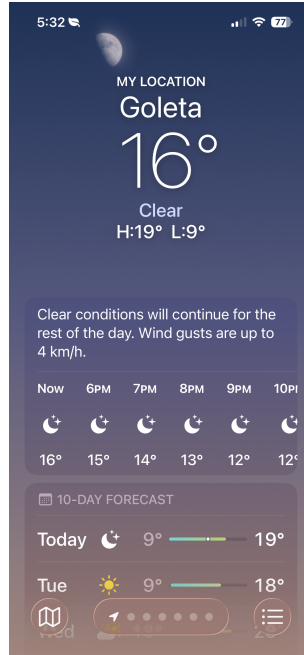


Figure 1: A screenshot of the temperature during the lab (Monday, January 26, 2026)

For the molecular gas mass, we'll accept the value $m \approx 4.81 \cdot 10^{-26} \text{ kg}$, which is derived from the weighted average of the masses of the common gases based on their percentage within the atmosphere. The molecular gas masses of the common gases, are found on the National Institute of Standards and Technology website.

(Source: National Institute of Standards and Technology, "Search for Species Data by Molecular Weight".<https://webbook.nist.gov/>)

2.2 Sources of Deviation in γ

If consider the potential inaccuracy of γ , the humidity of the day may be a potential source of error: As the air gets more humid, the water molecules in it would increase the overall mass of the gas, hence increasing the density. Yet, the constant γ is also related to the density by the formula $B = \rho \frac{dP}{d\rho}$. Which, with the lab day being somewhat humid, the constant γ might not accurately be 1.4.

2.3 Uncertainties

For k_B the Boltzmann constant, the uncertainty is assumed to be 0 (since according to the instruction one can believe it's near an exact value). For γ , one could suppose $\delta\gamma = \pm 0.05$ (since the given value is up to the 10th decimal place, which the error by 0.05 is not possible to verify). Finally, for the temperature, one has $\delta T = \pm 0.5$ K (since the temperature given by the weath report is always in integer values of unit Celcius, so an error of 0.5 K is not possible to be observed).

2.4 Proposed Speed of Sound

Given the values, one has the following value for the speed of sound:

$$v = \sqrt{\frac{\gamma k_B T}{m}} = \sqrt{\frac{1.4 \cdot 1.380649 \cdot 10^{-23} \cdot 289.15}{4.81 \cdot 10^{-26}}} \approx 340.875 \text{ m/s} \quad (6)$$

Which, the unit of the middle expression is $\sqrt{\frac{\text{J}}{\text{kg}}} = \sqrt{\frac{\text{kg} \cdot \text{m}^2/\text{s}^2}{\text{kg}}} = \sqrt{\frac{\text{m}^2}{\text{s}^2}} = \text{m/s}$.

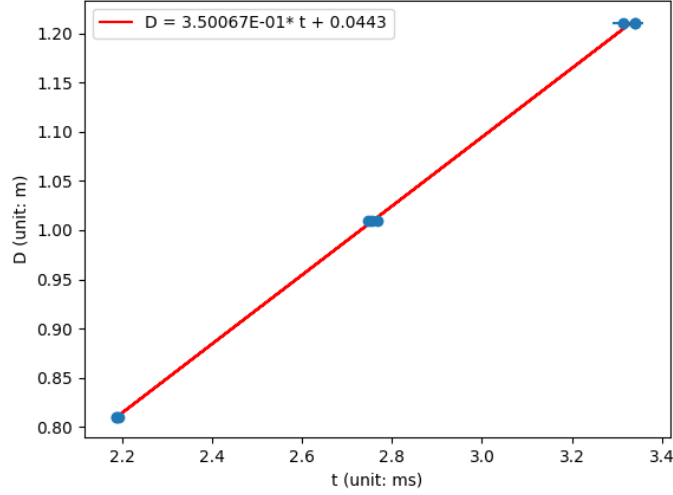
3 Data Analysis

For both approaches (with different sound frequency), we had gathered the average time difference and time difference uncertainty for each trial. Which, D (unit: m) represents the distance of the microphone from the speaker, t_i (unit: ms) represents the average time difference between the original and the received signal in the i th trial.

3.1 Frequency $f = 1$ kHz

The data is given in the following table: When making into scatter plot, one yields the following:

Distance (m)		Time (ms)					
D	δD	t_1	δt_1	t_2	δt_2	t_3	δt_3
0.81	0.0005	3.3133	0.0231	3.3400	0.0200	3.3400	0.0000
1.01	0.0005	2.7667	0.0115	2.7467	0.0115	2.7533	0.0115
1.21	0.0005	2.1867	0.0115	2.1933	0.0115	2.1867	0.0115



So, the best-fitting line for the data is given as follow:

$$D = 3.50067 \cdot 10^{-1}t + 0.0443 \quad (7)$$

Where D is the distance the wave traveled (unit: m), and t is the time it takes for the travel (unit: ms).

To calculate the uncertainties of the parameters, for the slope it'll use the formula as follow:

$$\delta_m = m \frac{\sqrt{1/R^2 - 1}}{\sqrt{n - 2}} \quad (8)$$

where R^2 represents the coefficient of determination, $R = \sum(y_i - y(x_i))^2$. On the other hand, the intercept uncertainty is given by:

$$\delta_k = \delta_m \sqrt{\frac{\sum x_i^2}{n}} \quad (9)$$

Plugin the values, one yields the following:

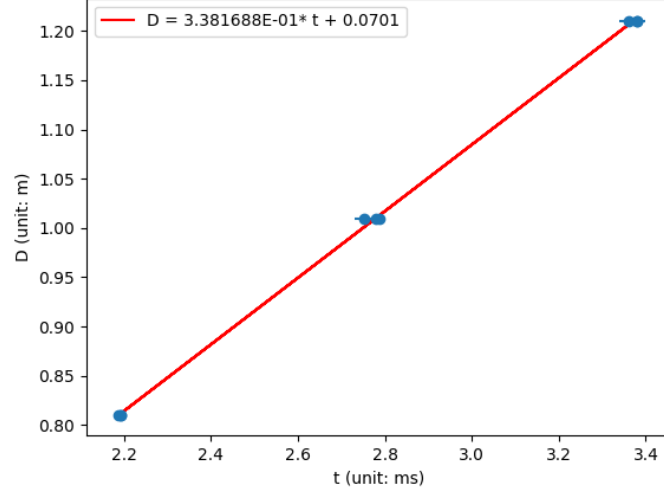
$$R^2 = 0.9996, \quad \delta_m \approx 0.0026467 \text{ m/ms}, \quad \delta_k \approx 0.0074 \text{ m} \quad (10)$$

As a result, the best-fit speed of sound in air would be given by the slope of the best-fit line, which is $3.50067 \cdot 10^{-1} \text{ m/ms}$, which translates to speed $v = 350.0670 \text{ m/s}$. And, since v is obtained by $1000 \cdot m$ (where m is the slope of the line of best fit), then its propagation of uncertainty $\delta v = 1000 \cdot \delta_m \approx 2.6467 \text{ m/s}$.

3.2 Frequency $f = 1.25 \text{ kHz}$

The data is given in the following table: When making into scatter plot, one yields the following:

Distance (m)		Time (ms)					
D	δD	t_1	δt_1	t_2	δt_2	t_3	δt_3
1.21	0.0005	3.3600	0.0200	3.3800	0.0200	3.3800	0.0200
1.01	0.0005	2.7867	0.0115	2.7533	0.0231	2.7800	0.0200
0.81	0.0005	2.1867	0.0115	2.1933	0.0115	2.1933	0.0115



So, the best-fitting line for the data is given as follow:

$$D = 3.381688 \cdot 10^{-1}t + 0.0701 \quad (11)$$

Using the formulas mentioned in the previous section, one yields the following for uncertainties of slopes and intercepts:

$$R^2 \approx 0.9995, \quad \delta_m \approx 0.0028588 \text{ m/ms}, \quad \delta_k \approx 0.0081 \text{ m} \quad (12)$$

Similar to the previous part, the best-fit speed of sound in air would be given by the slope of the best-fit line, which is $3.381688 \cdot 10^{-1} \text{ m/ms}$, which translates to speed $v = 338.1688 \text{ m/s}$.

Again, similar to the previous part, the propagation of uncertainty $\delta v = 1000 \cdot \delta m \approx 2.8588 \text{ m/s}$.

3.3 T-Test

If performing T-test on the gathered data (which are the time differences), since the two data sets all have 9 data points, the T -value is given as follow:

$$T = \frac{t_{\text{ave},1} - t_{\text{ave},2}}{\sqrt{\frac{(n_1-1)\text{Var}(t_1)^2 + (n_2-1)\text{Var}(t_2)^2}{n_1+n_2-2} \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} \quad (13)$$

With $t_{\text{ave},1} \approx 2.7585 \text{ ms}$, $\text{Var}(t_1) \approx 0.2447 \text{ ms}^2$, $t_{\text{ave},2} \approx 2.7793 \text{ ms}$, $\text{Var}(t_2) \approx 0.2622 \text{ ms}^2$, and $n_1 = n_2 = 9$, one yields $T \approx -0.0067$. If chosen the significance level $\alpha = 0.05$, one can see the T -value is significantly lower than α . Hence, the rejection failed, showing there's no evidence for significant difference between the two data sets.

4 Reflection

Based on the two sets of data collected (for $f = 1, 1.25$ kHz respectively), one yield the best-fit speed of sound being $v_1 = 350.0670$ and $v_2 = 338.1688$ m/s respectively, with uncertainty $\delta v_1 = 2.6467$ and $\delta v_2 = 2.8588$ m/s.

In comparison to the model's prediction $v = 340.8750$ m/s, the difference are between 2 to 10 m/s, which is somewhere between 1 to 5 scale of uncertainty away from the prediction. Looking at the percent error, one yields 2.6258% and 0.8003% respectively. Which, one has a high confidence to conclude that the measurements agree with the model prediction in this lab.

Finally, about some potential improvements of the lab:

- First, since the microphone has the receiver being a solid ball, one can't directly tell the precise position where the microphone received the signal. As a solution (which is mentioned in the In Lab), instead of using the absolute position of the microphone, using the relative position between different positions people placed it, and consider the new time difference as the difference between two time data points. This will avoid the situation where the receiver of the microphone is unclear. To compensate for this change, one needs to increase more position points to be measured.
- Another interesting phenomenon that appeared in the data set, is as the frequency gets higher, the measured time difference has an increasing trend. This suggests there are also some other external factors that may be affecting the result. To compensate for that, there's also a necessity of increasing trials regarding the frequency of the sound waves.